



BANK LOAN ANALYTICS

Group- 1

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- Dashboard Overview
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List of KPI's

1, Year Wise Loan Amount Analysis

- **Problem:**

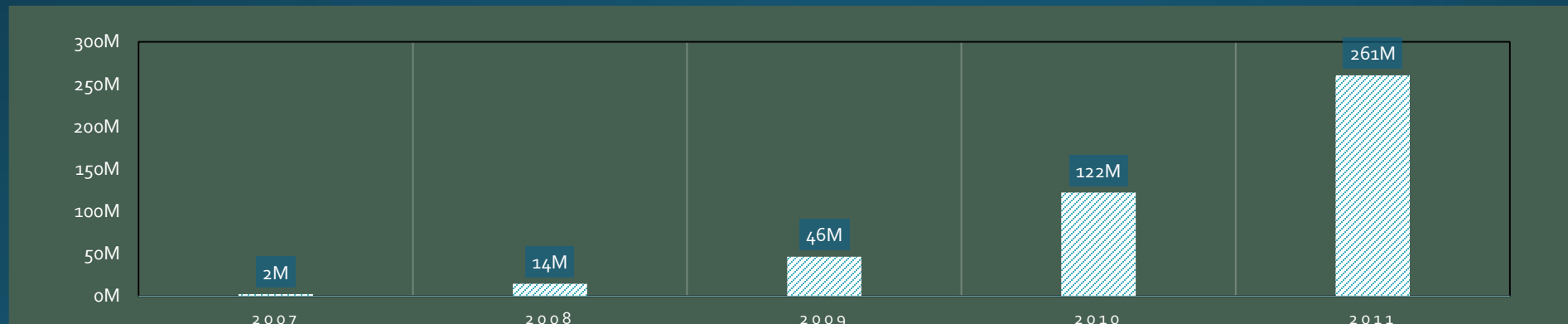
Analyze yearly loan distribution and growth trends.

- **Solution:**

Calculated total loan amount issued year-wise.

- **Insight:**

Loan amount increased significantly in later years, especially during 2011.



List of KPI's

Grade & Sub Grade Wise Revolving Balance

- **Problem:**

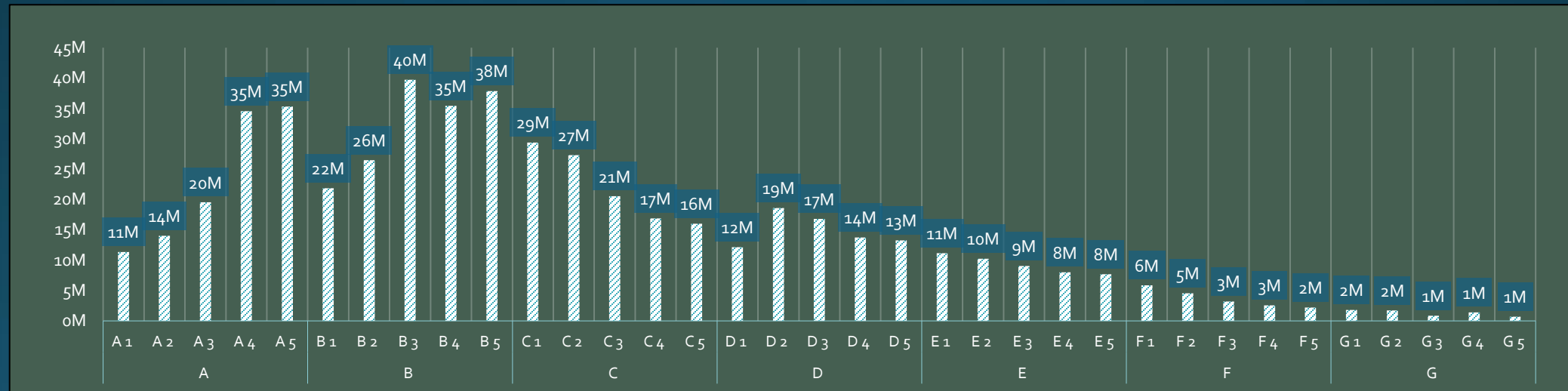
Identify revolving balance across customer grades.

- **Solution:**

Analyzed revolving balance using grade and sub-grade categories.

- **Insight:**

B and C grade customers showed higher revolving balances



List of KPI's

Verified vs Non-Verified Payment Analysis

- **Problem:**

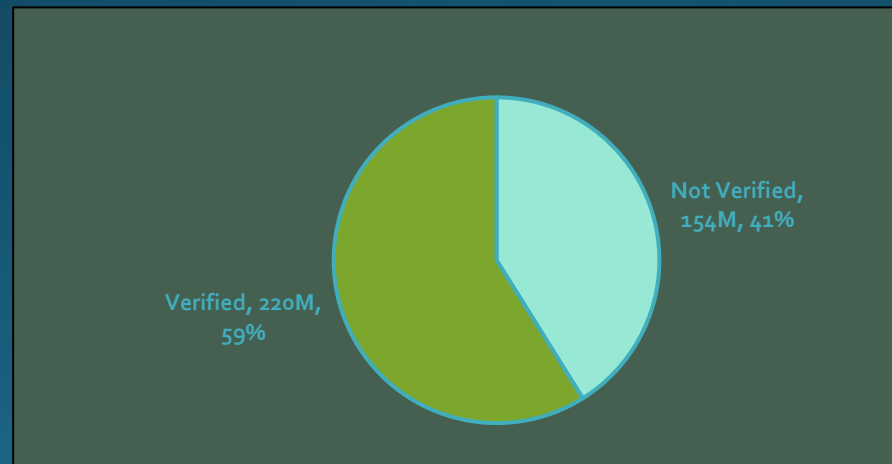
Compare payment behavior between verified and non-verified customers.

- **Solution:**

Calculated total payments based on verification status.

- **Insight:**

Verified customers contributed higher payments compared to non-verified customers



List of KPI's

State & Month Wise Loan Status Analysis

- **Problem:**

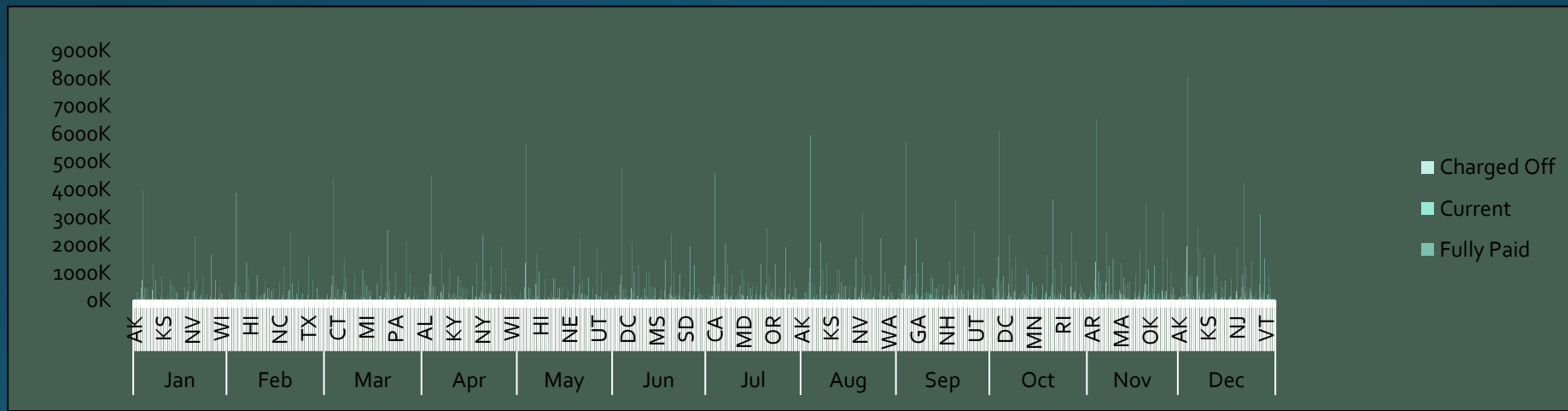
Analyze loan trends across different states and months.

- **Solution:**

Compared loan status state-wise and month-wise.

- **Insight:**

States like California and New York had higher loan activities



List of KPI's

5. Home Ownership vs Last Payment Analysis

- **Problem:**

Understand payment trends based on home ownership type.

- **Solution**

Compared payments across RENT, OWN, and MORTGAGE categories.

- **Insight:**

Mortgage customers contributed higher payment amounts.



EXCEL- Dashboard

BANK LOAN OF CUSTOMERS DASHBOARD

issue_d(Ye...

- 2007
- 2008
- 2009
- 2010
- 2011

home_own...

- MORTGAGE
- NONE
- OTHER
- OWN
- RENT

grade

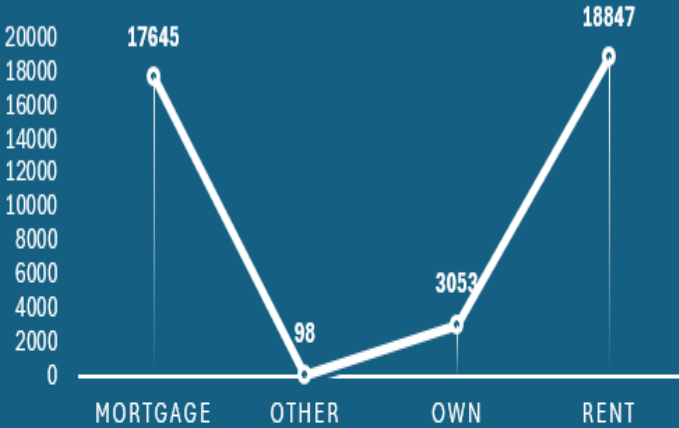
- A
- B
- C
- D

Total Loan Amount
445.6M

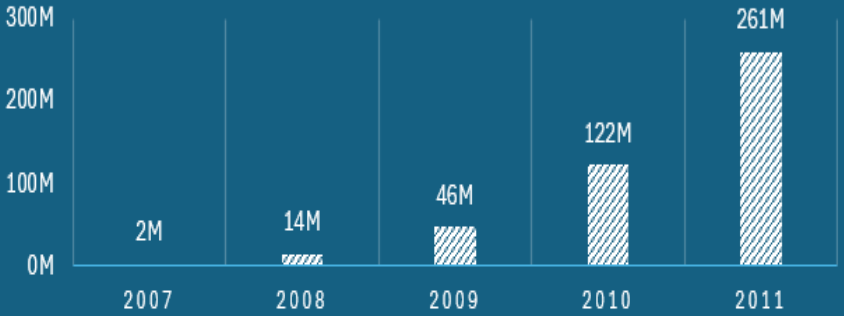
Total Payment Amount
482.70M

Unique_ID
39717

Count of last_pymnt_d



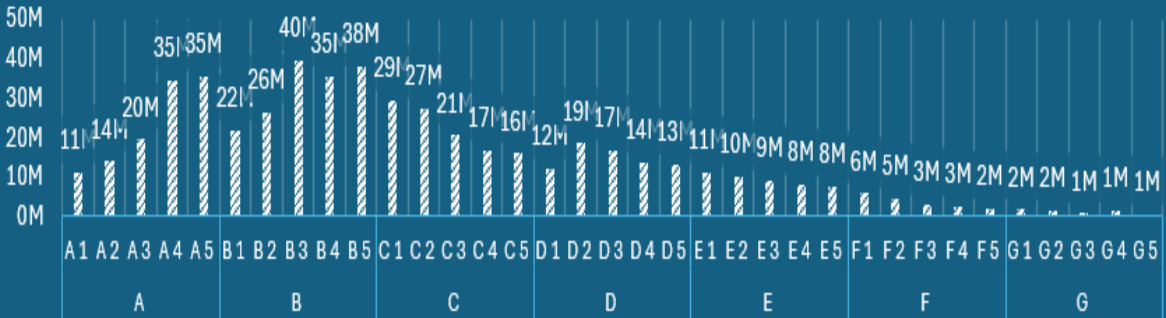
Sum of loan_amnt



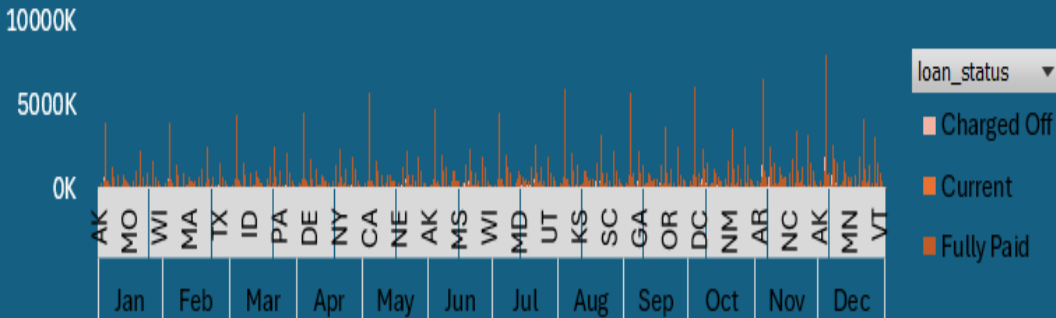
Sum of total_pymnt



Sum of revol_bal



Sum of loan_amnt



loan_status

- Charged Off
- Current
- Fully Paid

SQL WORK

```
1
2      #File type:  CSV UTF-8 (Comma delimited)
3
4 • CREATE DATABASE BANK_ANALYSIS;
5
6 • USE BANK_ANALYSIS;
7
8 • SELECT *FROM FINANCE1;
9 • SELECT *FROM FINANCE2;
10
11 • ALTER TABLE FINANCE1 CHANGE COLUMN `id` ID INT;
12 • ALTER TABLE FINANCE2 CHANGE COLUMN `id` ID INT;
13
14 • SET SQL_SAFE_UPDATES = 0;
15 |
16 • UPDATE finance1
17   SET issue_d = STR_TO_DATE(issue_d, '%d-%m-%Y');
18
19 • ALTER TABLE finance1
20   MODIFY COLUMN issue_d DATE;
21
22 • DESC finance1;
23 • DESC finance2;
24
25 -- KPI 1 : Year Wise Loan Amount Stats
26
27 • SELECT
28     YEAR(issue_d) AS year,
29     CONCAT(ROUND(SUM(loan_amnt) / 1000000, 0), 'M') AS total_loan
30 FROM finance1
31 GROUP BY year
32 ORDER BY year;
33
34 -- KPI 2 : Grade and Sub Grade Wise Revolving Balance
35
36 • SELECT f1.grade,f1.sub_grade,
37         ROUND(SUM(f2.revol_bal)/1000000,2) AS revolving_balance_million
38 FROM finance1 f1
39 JOIN finance2 f2
40 ON f1.id = f2.id
41 GROUP BY f1.grade,f1.sub_grade
42 ORDER BY f1.grade,f1.sub_grade;
43
```

```
44 -- KPI 3 : Verified Status vs Non Verified Total Payment
45
46 • SELECT f1.verification_status,
47         CONCAT(ROUND(SUM(f2.total_pymnt)/1000000,0),'M') as total_payment
48 FROM finance1 f1
49 JOIN finance2 f2
50 ON f1.id=f2.id
51 WHERE f1.verification_status in
52 ('verified', 'not verified') GROUP BY 1;
53
54
55 -- KPI 4 : State Wise and Month Wise Loan Status
56
57 • select addr_state as state,
58        monthname(issue_d) month ,loan_status,
59        concat(round(sum(loan_amnt)/1000000,0),'M')
60        total_loan from finance1
61 group by state,month, loan_status;
62
63 -- KPI 5 : Home Ownership vs Last Payment Date Stats
64
65 • SELECT f1.home_ownership,
66         COUNT(f2.last_pymnt_d) as payment_count,
67         CONCAT(ROUND(SUM(f2.last_pymnt_amnt)/1000,0),'K') as total_last_payment
68 FROM finance1 f1
69 JOIN finance2 f2
70 ON f1.id=f2.id
71 GROUP BY f1.home_ownership
72 ORDER BY total_last_payment DESC;
```

Result Grid		
	year	total_loan
▶	2007	2M
	2008	14M
	2009	46M
	2010	122M
	2011	261M

Result Grid		
	verification_status	total_payment
▶	Verified	220M
	Not Verified	154M

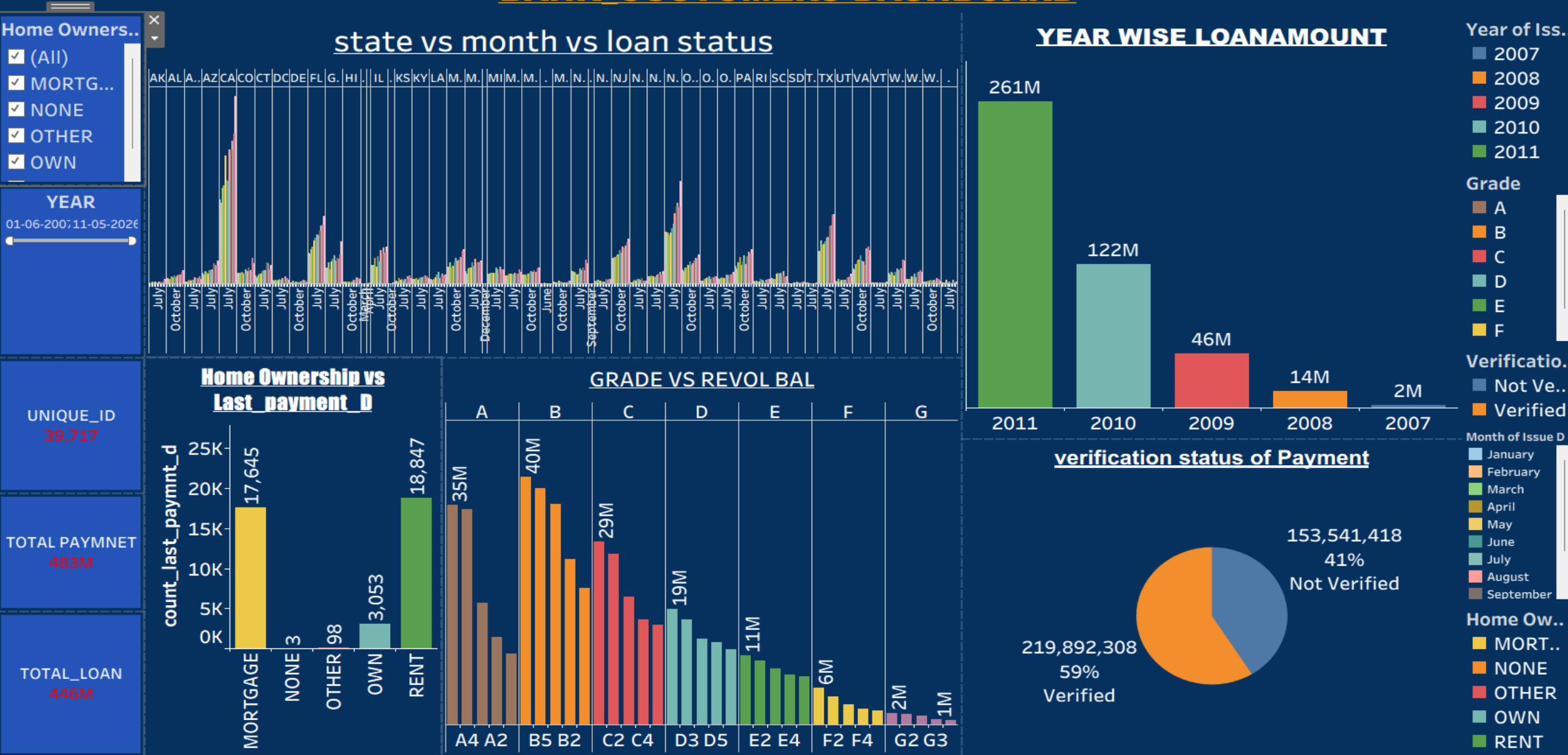
Result Grid			
	home_ownership	payment_count	total_last_payment
▶	OWN	3058	8229K
	MORTGAGE	17659	56306K
	RENT	18899	41687K
	NONE	3	1K
	OTHER	98	173K

Result Grid		
	grade	sub_grade
▶	A	A1
	A	A2
	A	A3
	A	A4
	A	A5
	B	B1
	B	B2
	B	B3
	B	B4
	B	B5
	C	C1
	C	C2
	C	C3
	C	C4
	C	C5
	D	D1
	D	D2
	D	D3
	D	D4
	D	D5
	E	E1
	E	E2
	E	E3
	E	E4
	E	E5
	F	F1

Result Grid				
	state	month	loan_status	total_loan
▶	CA	August	Fully Paid	6M
	NY	May	Fully Paid	2M
	TX	March	Fully Paid	2M
	CT	November	Fully Paid	1M
	CA	April	Fully Paid	5M
	TX	September	Fully Paid	3M
	TX	February	Fully Paid	2M
	MD	February	Charged Off	0M
	MA	April	Fully Paid	1M
	MI	August	Charged Off	0M
	CA	May	Fully Paid	6M
	KY	February	Fully Paid	0M
	MS	December	Fully Paid	0M
	KS	June	Fully Paid	0M
	MI	October	Fully Paid	1M
	GA	February	Fully Paid	1M
	CA	March	Fully Paid	4M
	NC	March	Fully Paid	1M
	NJ	August	Fully Paid	2M
	NY	September	Fully Paid	4M
	WA	March	Fully Paid	0M
	KY	April	Fully Paid	0M
	OH	May	Fully Paid	1M
	AZ	May	Fully Paid	0M
	MI	March	Fully Paid	1M
	CT	February	Fully Paid	0M

TABLEAU DASHBOARD

BANK CUSTOMERS DASHBOARD



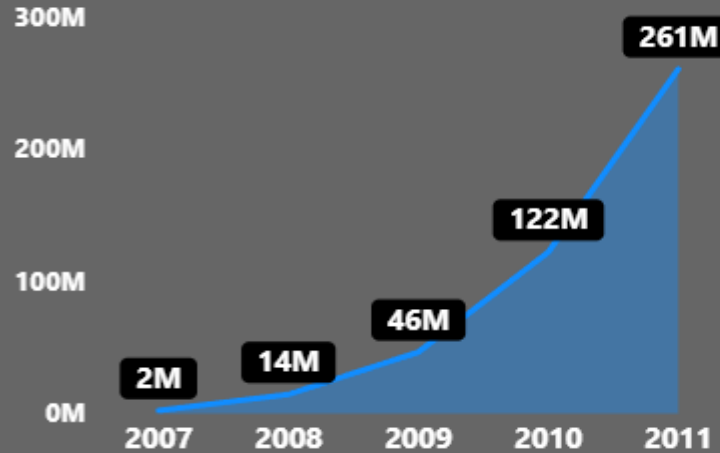
POWER BI DASHBOARD

Clear all slicers

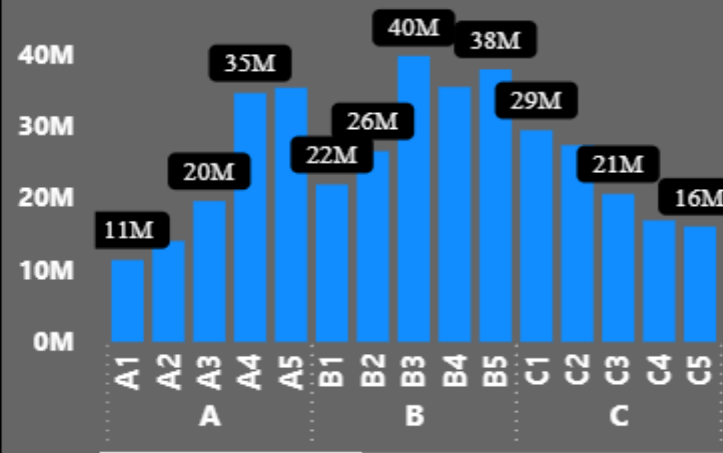
BANK CUSTOMER DASHBOARD



Sum of loan_amnt by Year



Sum of revol_bal by grade and sub_grade



446M

Sum of loan_amnt

482.70M

Sum of total_pymnt

Year

- ☐ (Blank)
- ☐ 2008
- ☐ 2009
- ☐ 2010
- ☐ 2011
- ☐ 2012
- ☐ 2013
- ☐ 2014

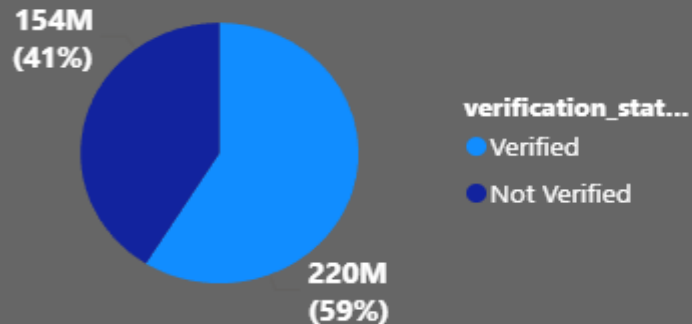
grade

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E
- ☐ F
- ☐ G

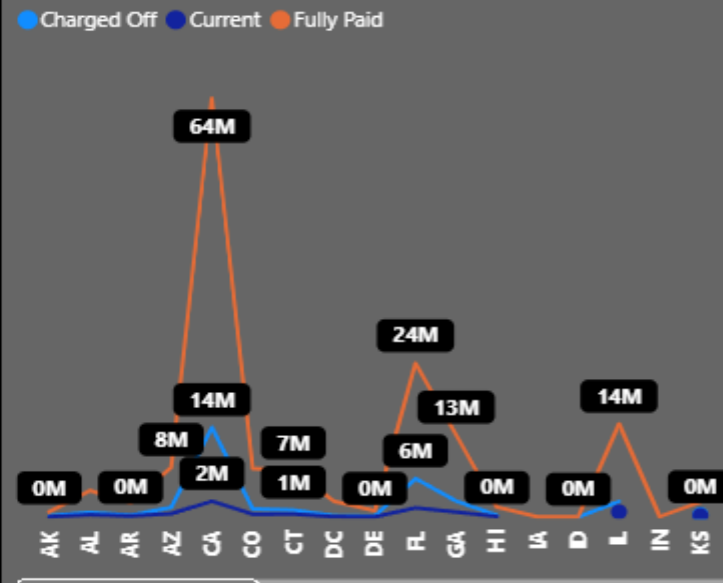
home_...

- ☐ MORT...
- ☐ NONE
- ☐ OTHER
- ☐ OWN
- ☐ RENT

Sum of total_pymnt by verification_status

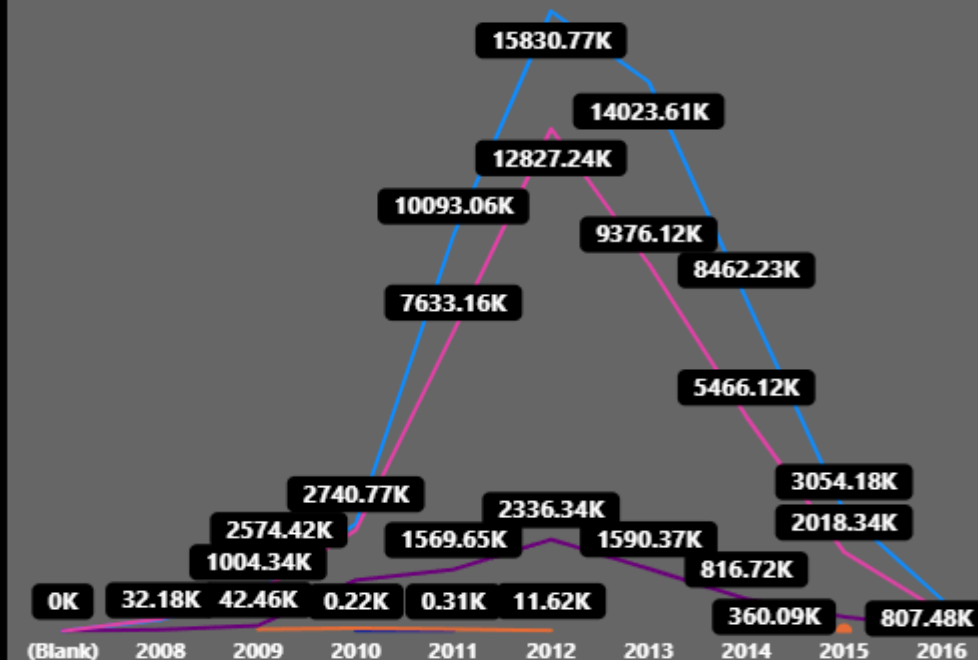


Sum of loan_amnt by addr_state and loan_status



Sum of last_pymnt_amnt by Year and home_ownership

MORTGAGE NONE OTHER OWN RENT



PROBLEMS IDENTIFIED

- ✓ Loan growth trends were unclear.
- ✓ Customer payment behavior needed analysis.
- ✓ High-risk customer grades were not identified.
- ✓ State-wise loan performance lacked visibility.
- ✓ Payment trends based on home ownership were unknown

SOLUTION APPROACH

- Cleaned and prepared finance datasets.
- Performed KPI analysis using SQL.
- Created relationships between tables.
- Built interactive Tableau dashboard.
- Used filters and visualizations for better analysis.

RECOMMENDATIONS

- ✓ Focus more on verified customers.
- ✓ Monitor high revolving balance customers.
- ✓ Improve loan monitoring in high-risk states.
- ✓ Enhance payment tracking systems.
- ✓ Use dashboard insights for financial decision-making

CONCLUSION

- Loan demand increased over the years.
- Verified customers showed better payment behavior.
- Certain customer grades had higher revolving balances.
- Tableau dashboard provided interactive business insights.
- Data analytics helps improve financial decision-making



Thank You